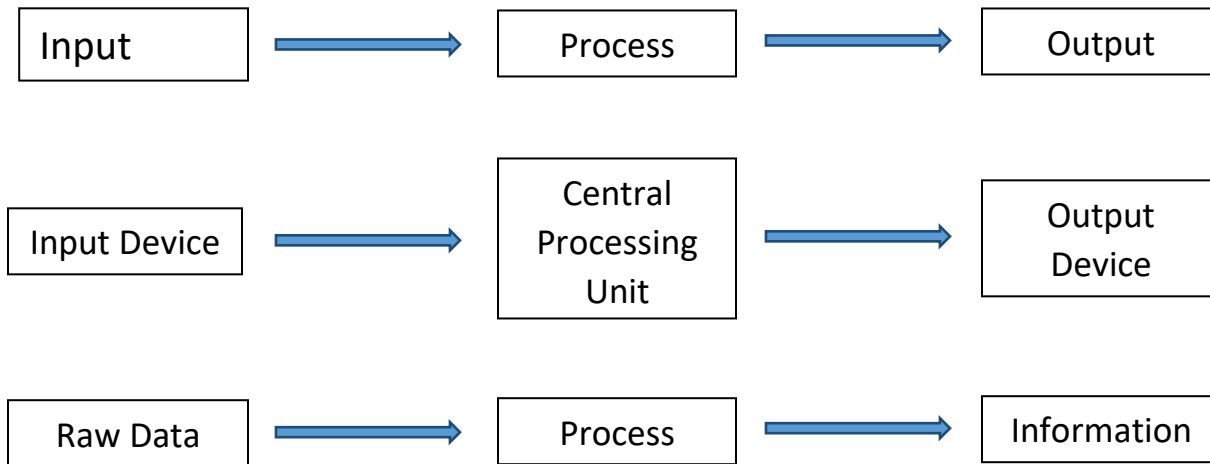


Introduction to Computer and Operating System

What is a Computer?

Computer is an electronic device designed to accept raw data; perform prescribed mathematical and logical operations at high speed and display the results of these operations.

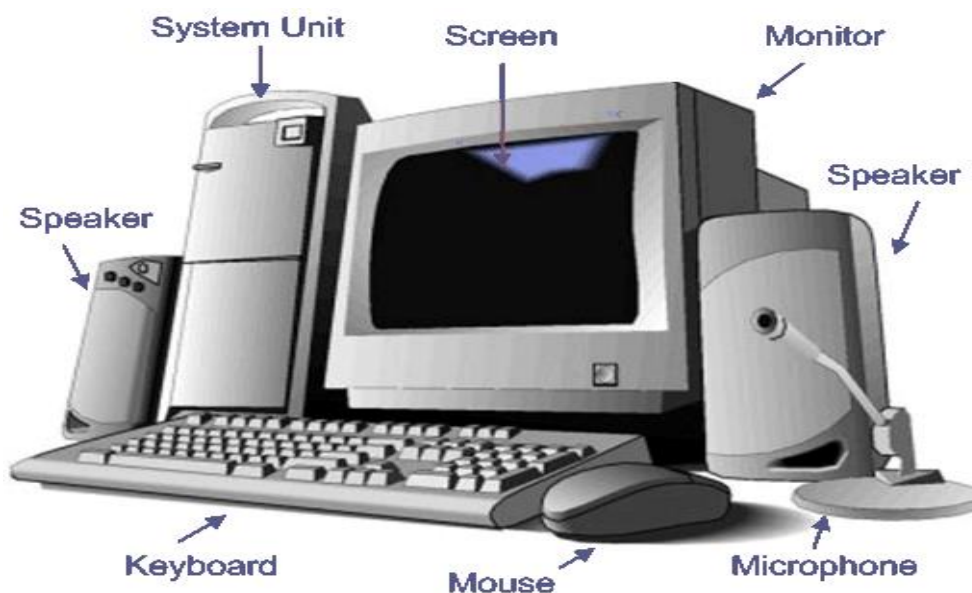


Main Components of a Computer:

In general, a computer system is composed of three elements.

- ☐ Hardware - The physical (electronic and mechanical parts) components of the computer.
- ☐ Software - Data and programs (Applications) run on the computer.
- ☐ Liveware - The user who uses the computer.

Parts of a computer:



Input Devices:

An **input device** is any hardware device that sends data to a computer, allowing you to interact with and control it. Examples for Input Devices;

- | | | |
|-------------------------------------|---|--|
| ☐ Keyboard | ☐ Scanner | ☐ Optical Character Reader(OCR) |
| ☐ Mouse | ☐ Graphic Tablet | ☐ Bar Code Reader |
| ☐ Joy Stick | ☐ Microphone | ☐ Optical Mark Reader(OMR) |
| ☐ Light pen | ☐ Magnetic Ink Card Reader(MICR) | |
| ☐ Track Ball | | |

Output Devices:

Output devices are used to carry the information from your computer and display the information on paper, on the screen or through the speakers. Examples for output devices;

- | | | |
|---|---|--------------------------------------|
| ☐ Monitors | ☐ Projector. | ☐ Speaker(s) |
| ☐ Printers (all types) | ☐ LCD Projection Panels. | ☐ Head Phone. |
| ☐ Plotters. | | |

Hardware Components:

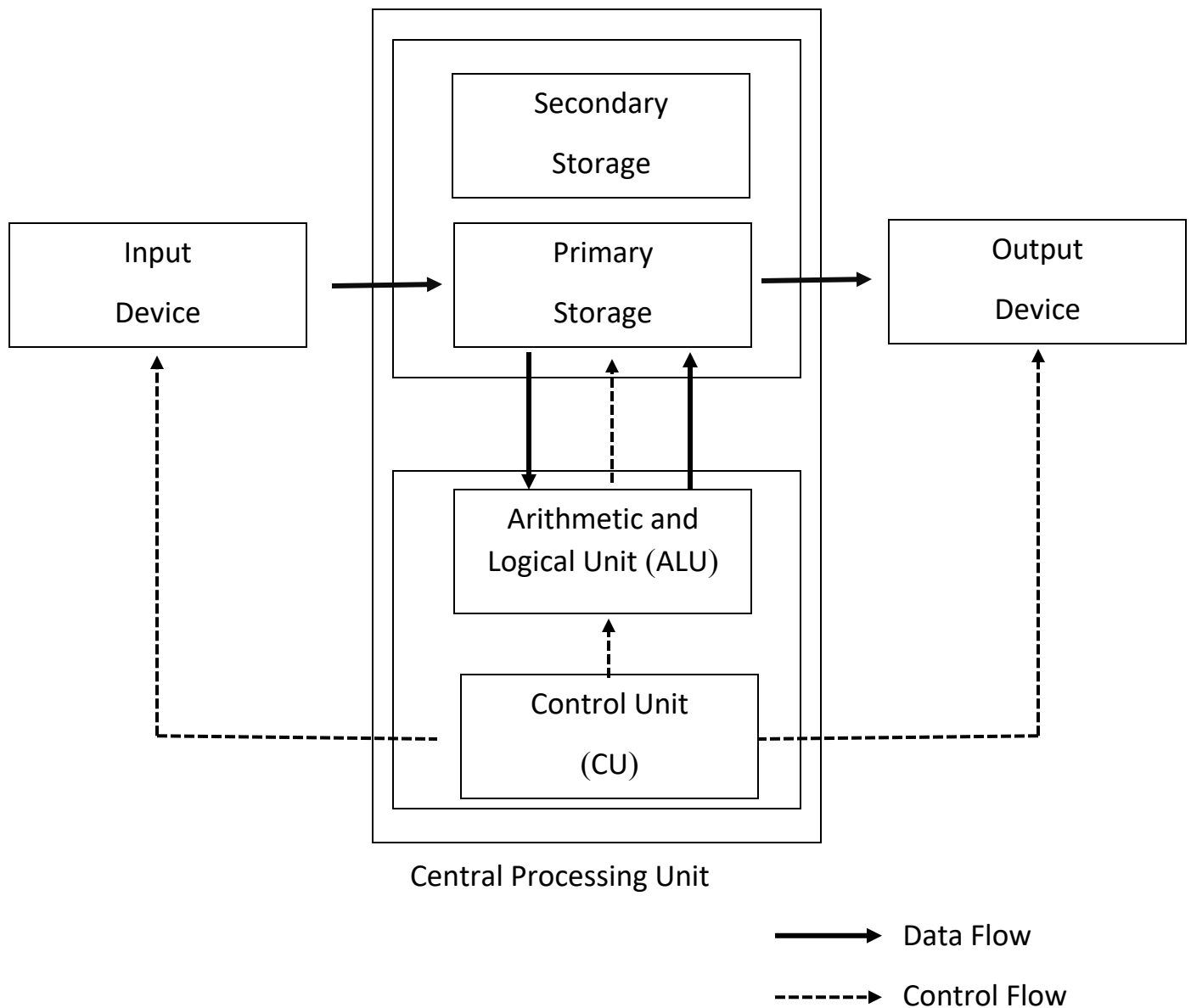
All hardware components are divided into three main parts.

1. Central Processing Unit
2. Storage Unit
3. I/O Devices

1. Central Processing Unit is again divided into two parts;
 - i. Arithmetic and Logical Unit (ALU)
 - ii. Control Unit (CU)

2. Storage Unit is again divided into two main parts
 - i. Primary Storage
 - ii. Secondary Storage

3. I/O Devices are Input Devices and Output Devices.



Computer Memory:

Computer **memory** is any physical device capable of storing information temporarily or permanently. The memory of the computer is divided into two categories:

1. Primary Memory
2. Secondary Memory

Primary Memory

This is the main memory of the computer. CPU can directly read or write on this memory. It is fixed on the motherboard of the computer. Primary memory is further divided in two types:

1. **RAM(Random Access Memory)**

RAM is a temporary memory. The information stored in this memory is lost as the power

supply to the computer is turned off. That's why it is also called **Volatile Memory**. It stores the data and instruction given by the user and also the results produced by the computer temporarily.

2. ROM(Read Only Memory)

Information stored in ROM is permanent in nature, i.e., it holds the data even if the system is switched off. It holds the starting instructions for the computer. ROM cannot be overwritten by the computer. It is also called Non-Volatile Memory. Its contents are written at the time of manufacture, but in modern computers may be changed using special software.

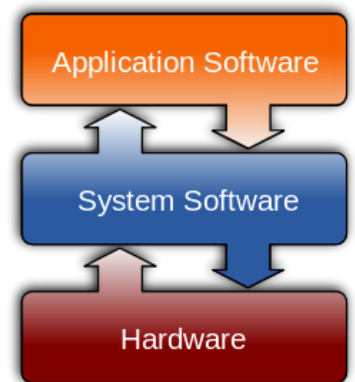
Secondary Memory

This memory is permanent in nature. It is used to store the different programs and the information permanently (which were temporarily stored in RAM). It holds the information till we erase it. Examples for secondary storage devices;

- ☐ Hard Disc
- ☐ Compact Disc
- ☐ DVD
- ☐ Pen Drive
- ☐ Blue Ray Disk

Software:

A software is a means of communication between the computer system and computer users. It is the operating system and applications that are used in computers. Basically, it is the collection of computer programs, documentation and procedures performing several tasks on a computer system. Without software, most hardware would sit there doing nothing or perform specific tasks. To make most hardware run we need to use software, and your task here is to select the correct type of software for each job. The two main classifications of software that all programs fit under are:



- ☐ System software
- ☐ Application software

System software –

System software is designed to **operate the computer hardware and to provide a platform for running application software**. Modern computers are complex machines involving many different parts. To keep it running well you will need system software. System software will handle the smooth running of all the components of the computer as well as providing general functionality for other programs to use, tools to speed up the computer, tools to develop new

software and programs to keep you safe from attacks. There are several different types of system software.

☐ **Operating Systems**

This is a collection of programs that make the computer hardware conveniently available to the user and also hide the complexities of the computer's operation. The Operating System interprets commands issued by application software (e.g. word processor and spreadsheets). The Operating System is also an interface between the application software and computer.

Below are some of its functions.

- ☐ Provide security and backup.
- ☐ Booting computers.
- ☐ Controlling peripheral devices such as a keyboard, disk drives, printers, etc.
- ☐ Provide interface between software and hardware.
- ☐ Scheduling processes.
- ☐ Memory management.
- ☐ Keeping track of files and directories on the disk.
- ☐ Recognize input from the keyboard.
- ☐ Send output to the display screen.

☐ **Utility programs**

These are small, powerful programs with a limited capability, they are usually operated by the user to maintain a smooth running of the computer system. Examples;

- ☐ file management
- ☐ diagnosing problems and finding out information about the computer
- ☐ copy, paste, delete,
- ☐ file searching
- ☐ disk defragmenter
- ☐ disk cleanup.

☐ **Library programs**

These are a compiled collection of subroutines (e.g. libraries make many functions and procedures available when you write a program)

☐ **Translator software**

- ☐ **Assembler** translates assembly language programs into machine code (A binary code that a machine can understand).
- ☐ **Compiler** translates high level language code into object code; which is the machine language of the target machine.
- ☐ **Interpreter** analyses and executes a high-level language program a line at a time. Execution will be slower than for the equivalent compiled code as the source code is analyzed line by line.

Application software –

Application software is designed for people to perform tasks that we consider useful. There are several categories of Application software.

□ General purpose application software.

Software is general-purpose if it can be used for lots of different tasks. Examples;

- ☐ Word Processing Software
- ☐ Database Software
- ☐ Spreadsheet Software
- ☐ Multimedia Software
- ☐ Presentation Software

□ Special purpose application software.

Special purpose software is also called Tailored software. Tailored software is for specific purposes. Banking software, hotel reservation software, hospital software, billing software, etc. are its examples.

□ Bespoke application software

Bespoke software is written for a single client. Large organizations have a need for well-developed applications suited to their specific needs. Such software is often expensive to develop since the development costs are not shared among a large number of people purchasing the software.

Characteristics of Computer:

- ☐ **Efficiency:** The *computer* processed data in Nano second which is beyond of human capacity.
- ☐ **Speed:** It is with very high processing speed. The computer can process more than 50 million instructions in one second. Now a day, very high speed computers are available with Pentium IV or Pentium v microprocessor chips.
- ☐ **Storage Capacity:** It can store vast amount of information in the form of books, graphics, charts, animation, audio, video etc. Now a day, the hard disk of 1024 GB is common, and hard disk of more than 1 TB is also available in market.
- ☐ **Versatility:** It has capacity of multitasking. You can enjoy multimedia, word processing, printing, and communicating jobs from one platform without closing other. This type of facilities has made it versatile and popular.
- ☐ **Reliability:** The computer is a reliable electronic multipurpose and multiprocessing machine. It processes data with high accuracy without any mistakes.
- ☐ **Diligence:** - A computer is free from tiredness, lack of concentration, fatigue, etc. It can work for hours without creating any error. If millions of calculations are to be performed, a

computer will perform every calculation with the same accuracy. Due to this capability it overpowers human being in routine type of work.

- ☐ **Decision Making:** The computer has capability to perform task and managing processing schedules when there are many jobs available and Waiting own turns. The Operating system has capability to manage files, executing utilities software and many others. The game, weather forecasting, weaponry systems, aviation's, space shuttles etc. are controlled by computer and the controlling actions are performed by its decision making capacity based on stored related database.
- ☐ **Accuracy:** The computer is applied in scientific and industrial calculation where long range data is used for information.
- ☐ **Artificial Intelligence:** The fifth generation computer which recognizes voice is based on artificial intelligence. The AI is applied in gaming software, and robotics from many years, but operating system based on AI is also in process which can recognize not only voice, but read mood of users also. It gives suggestion and decision on complex problems.

Applications of Computer:

The various applications of computers in today's arena :

1. Business

- ☐ Payroll calculations ☐ Budgeting ☐ Managing employee's database
- ☐ Sales analysis ☐ Financial forecasting ☐ Maintenance of stocks

2. Education

- ☐ Source of information ☐ Online Universities
- ☐ E-books ☐ Video-based classes

3. Marketing

- ☐ Advertising ☐ Home Shopping

4. Banking

- ☐ Online accounting facility ☐ ATM machines

5. Insurance

- ☐ Procedure to continue with policies ☐ Survival benefits
- ☐ Handling Interests ☐ Bonus

6. Communication

- ☐ E-mail
- ☐ Chatting

- ☐ Video-conferencing

7. Health Care

- ☐ Diagnostic System
- ☐ Lab-diagnostic System
- ☐ Patient Monitoring System

- ☐ Pharma Information System
- ☐ Surgery

8. Military

- ☐ Missile Control
- ☐ Military Communication

- ☐ Military Operation and Planning
- ☐ Smart Weapons

9. Engineering Design

- ☐ Structural Engineering
- ☐ Industrial Engineering

- ☐ Architectural Engineering

10. Government

- ☐ Budgets
- ☐ Income tax department
- ☐ Computerization of voters lists
- ☐ Sales tax department
- ☐ Computation of male/female ratio
- ☐ Weather forecasting

11. Travelling

- ☐ Online ticket reservations
- ☐ Online timetables
- ☐ Hiring private cabs

12. Entertainment

- ☐ Gaming software
- ☐ Online movies
- ☐ Online music

Classification of Computer:

Computers differ based on their data processing abilities. They are classified according to purpose, data handling and functionality.

According to purpose-

☐ General purpose computers

These are designed to perform a range of tasks. They have the ability to store numerous programs, but lack in speed and efficiency.

☐ Specific purpose computers

These are designed to handle a specific problem or to perform a specific task. A set of instructions is built into the machine.

According to data handling-

According to data handling there are three types of computers.

- **Analog Computers**

An analog computer is a computer which is used to process analog data. Analog computers store data in a continuous form of physical quantities such as electrical, mechanical, or hydraulic quantities and perform calculations with the help of measures.

- **Digital Computers**

A computer that performs calculations and logical operations with quantities represented as digits, usually in the binary number system.

- **Hybrid Computer (Analog + Digital)**

A combination of computers those are capable of inputting and outputting in both digital and analog signals. A hybrid computer system setup offers a cost effective method of performing complex simulations.

According to the size of the Computer-

- **Super Computer**

The fastest and most powerful type of computer. Supercomputers are very expensive and are employed for specialized applications that require immense amounts of mathematical calculations. For example, weather forecasting requires a supercomputer. Other uses of supercomputers include animated graphics, nuclear energy research, and petroleum exploration.

- **Mainframe Computer**

A very large and expensive computer capable of supporting hundreds, or even thousands, of users simultaneously. In some ways, mainframes are more powerful than supercomputers because they support more simultaneous programs. But supercomputers can execute a single program faster than a mainframe.

- **Mini Computer**

A mid-sized computer. In size and power, minicomputers lie between workstations and mainframes. In general, a minicomputer is a multiprocessing system capable of supporting from 4 to about 200 users simultaneously.

- **Micro Computer or Personal Computer**

- **Desktop Computer:** A personal computer sufficient to fit on a desk.
- **Laptop Computer:** A portable computer complete with an integrated screen and keyboard. It is generally smaller in size than a desktop computer and larger than a notebook computer.
- **Palmtop Computer/Digital Diary /Notebook /PDAs:** a hand-sized computer. Palmtops have no keyboard but the screen serves both as an input and output device.

- **Workstations**

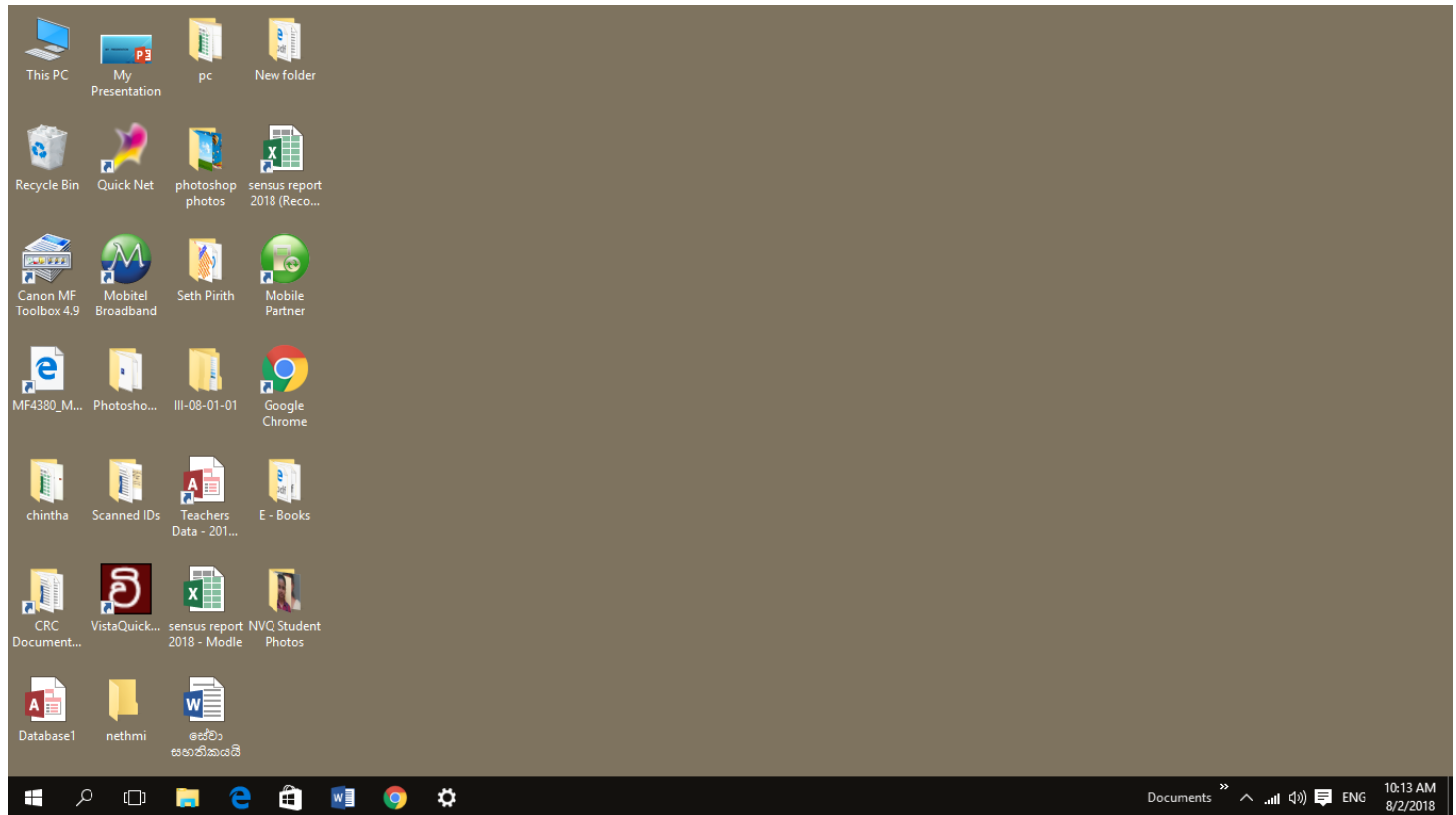
A terminal or desktop computer in a network.

Graphics Mode (Windows):

Graphics mode is a computer display mode that generates images using pixels. Today, most users operate their computer in a graphics mode opposed to a text mode or command line environment.

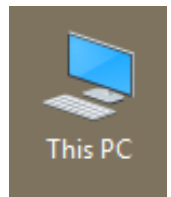
Active Desktop:

Active desktop makes it possible for you to customize your desktop, launch programs and switch between files.



Icons:

An Icon is a picture or symbol that represents a program, command feature, option or tool.



Task Bar:

A horizontal bar, located initially at the bottom of your screen that contains the Windows button or the Start button, clock, program icons and other icons.



When you open any program, the name of the open program goes to the Task bar as a tag. You don't have to close windows or move them aside to find other ones. Just click the button and the corresponding open program will become active.

Power Button:

When you click on the Windows button, you find the **Power** button. If you want to shut down the computer, click on **Shut down** or if you want to restart the computer, click on **Restart**.

Launching a Program:

1. Click on the Windows button → Click on the program you want
2. Click on the Windows button → Click on **All Apps** → Click on the program name you want
3. Click on the Search button on the task bar → Type the name of the program you want to open → Select the program from the given results
4. Right click on the Windows button → Run → Type the name of the exe file of the program you want to open or browse for the file → Ok

Finding Files and Folders:

Click on the “**Search Windows**” button on the Task Bar → Type the name of the file or folder to be searched → Click on the selected name of the file or folder.

Change the background of the Desktop:

Right click on the background → Personalize → Background → Select any option (Picture / Solid Color / Slideshow) → Select any picture / Color / Slideshow or browse for any other → If you select Slideshow, select other options as you want → Close the window

Change the Screen Saver:

Right click on the background → Personalize → Lock screen → Screen saver settings → Screen saver → Select any option → Set timing → If you want, click on “**Preview**” to preview the screen saver → Apply → Ok.

Task Bar Properties:

Right click on the Task Bar → Properties → Task Bar and Start Menu Properties → Use options in **Task Bar** and **Toolbars** tabs to do given changes.

“This PC” Icon:

The window of this icon displays the symbols for the physical content of your computer, including hard drive, CD ROM drive and other external storage devices.

“Recycle Bin” Icon:

The Recycle Bin is a temporary storage where old, deleted files from the hard drive hang out until you may need again. It can't be renamed or deleted. Files deleted using DOS programs are not sent to the recycle bin. Only deleted windows files are sent to it.

Sending a file / folder to the Recycle Bin –

Right click on the file / folder → Delete

Recovering the deleted file:

Open the Recycle Bin → Right click on the file / folder → Restore

Deleting the file permanently:

Open the Recycle Bin → Right click on the file / folder → Delete

File Name:

This is a collection of information stored on a computer with a unique name.

File Extension:

There are different types of files. To identify the file easily there is a tag called an extension.

Eg:

File Name	Extension	File Type
Clouds	.bmp	Bit map file
Scenery	.gif	Image file
Students	.docx	Word file
Salary	.xlsx	Excel file
My Show	.pptx	PowerPoint file

Understanding file / folder properties:

All files have properties or characteristics. Knowing about them makes our file management tasks quick and easy to use.

Right click on the relevant file / folder → Properties → It displays various tabs including full description about the file.

Creating Folders:

Right click on the desktop → **New** → **Folder** → Type a folder name → Enter

Renaming a file/folder:

Right click on the relevant file/folder → Rename → Type the new name → Enter

Moving and Copying files and folders:

There are three methods to move and copy.

1. Using Cut / Copy / Paste methods

- ☐ Using Menu commands
 - ☐ Using tool bar options
 - ☐ Using right click menu
 - ☐ Using keyboard
- Using any of the above methods,
- Select the file
 - Give the **Copy / Cut** command
 - Select the new location
 - Select the **Paste** command

2. Right drag and drop method

- ❑ Select the file that you want to move or copy → Right click the object and drag it to the new location → Release the mouse button → Select “**Move here**” / “**Copy here**”

3. Left drag and drop method

To Move –

Click on the object and drag it to the new location → Release the mouse button → The file is moved.

To Copy -

Click on the object and drag it to the new location while pressing the **ctrl** key → Release the mouse button → The file is copied.

Deleting files and folders:

Select the file to be deleted → Press **Delete** key / Right click on the selected file and select **Delete**.

“Undo” command:

When a particular command is given to do something and if you want to cancel the same command, you can cancel that by using the **Undo** command.

- ❑ Click **Undo** button/ Ctrl + Z / Right click → Undo

Creating a shortcut to a program on the desktop:

Right click on the object → Send to → Desktop (Create shortcut)

Creating a shortcut on the File Explorer:

Click on the application you want → Drag it onto the task bar → Release the mouse button → The shortcut will be created in File Explorer Quick Access.

Setting Mouse Pointers:

You can change one or more mouse pointers.

Control Panel → Open the **Mouse** icon →

- ❑ Use **Buttons** tab to switch between primary and secondary buttons and to change the double click speed
- ❑ Use **Pointers** tab to customize any type of mouse pointer and to enable pointer shadow
- ❑ Use **Pointer Options** tab to customize pointer options
- ❑ Use **Wheel** tab to customize the mouse wheel.

Adding Printers:

Control Panel → Devices and Printers → Add a Printer → The Printer that I want isn't listed → Add a local printer or network printer with manual settings → Next → Use an existing port → Next → Select the manufacturer → Select the Printer model → Next → If you want, change the printer name; otherwise remain it same → Next → If you want, set the printer as the default printer → If you want, print a test page → Finish → The printer will be added to the list.